

JAIDEEP VALLAPUNENI

New York City, NY | jai.fstec@gmail.com | +1 (410) 826-7008 | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Software Engineer with **3+ years of experience** building **scalable, high-performance distributed systems** using **Java/Spring Boot, Kafka, and React**, with a proven track record of **optimizing critical workflows** and delivering **reliable, impact-driven solutions for global scale**.

TECHNICAL SKILLS

- **Languages:** Java, Javascript, Python, Typescript, SQL, Shell script
- **Database:** Oracle, PostgreSQL, MongoDB
- **Frameworks & Cloud:** Spring, Spring boot, React, AWS
- **Build and deployment tools:** Gradle, Maven, Jenkins
- **Other:** HTML, CSS, Redis, REST, GraphQL, SOAP, Microservices, Splunk, EC2, S3, AWS Lambda, Snowflake, Kafka, ansible, Junit, Log4j, Jest, React Testing Library, Agile, Git, TDD, Docker, Kubernetes, JIRA, postman, Tomcat, Swagger

WORK EXPERIENCE

TD Bank, Full Stack Engineer

Jul 2024 – present | New York, USA

- Engineered and deployed **Java Spring Boot microservices** with **Apache Kafka** for event streaming, **RESTFUL** and **GraphQL** APIs for efficient data access, scaling to process millions of analyst events per day with sub-second latency.
- Led the **architectural integration** of React Micro-Frontends using Webpack **Module Federation** and a shared message bus, enabling independent deployment pipelines across **cross-functional teams**.
- Achieved **300 ms** query latency across millions of ESG records by integrating **ElasticSearch** with **Oracle**, enabling real-time KPI-driven decision making.
- Integrated **LLM capabilities** into the React dashboard to automate financial summarization, utilizing prompt engineering and context-aware APIs to reduce manual report drafting time.
- Reduced **API requests** by **60%** and improved **UI scalability and state consistency** by designing a configurable, reusable React component architecture with centralized state and routing (Redux Toolkit, Context API, Hooks, React Router).
- Increased **system availability by 10%** by developing a comprehensive disaster recovery plan, implementing automated infrastructure backups, and enhancing overall system resilience and fault tolerance.
- Configured **AWS API Gateway** and **Application Load Balancer** (ALB) to efficiently route and balance traffic across microservices, improving scalability and reducing response latency by 35%.
- Achieved a **30% increase** in user traffic by reducing UI load times through host-app caching, CDN integration, bundle compression, and component-level lazy loading.
- Developed end to end **MongoDB** aggregation and **ETL workflows** integrating with **Snowflake**, streamlining ESG data ingestion for backend services, reducing manual reconciliation by 90%.
- Conducted **anomaly detection** using Python based validation scripts, proactively identifying data quality issues and increasing reporting **accuracy by 98%**.

Lumen Technologies, Software Development Engineer

Aug 2021 – Jul 2022 | Bangalore, India

- Optimized a **Java Spring Boot** backend by refactoring synchronous payroll workflows into multi-threaded batch processes, reducing execution time by 75% and increasing throughput by 3x.
- Reduced REST API response time by **96% (3s to under 100ms)** by optimizing JPA/Hibernate mappings, adding DB indexes, implementing **Redis** caching, and enabling server-side sorting & pagination.
- Designed **React Application** that empowered support agents to triage customer tickets efficiently, cutting **resolution time by 40%** enhancing user experience and responsiveness.
- Modernized legacy payroll ingestion by architecting an event-driven serverless workflow using **AWS S3** triggers and **Lambda**, automating data enrichment against **Amazon RDS**.
- Implemented state management in React using **Redux**, improving application performance and maintainability.
- Orchestrated payroll microservices on **Kubernetes (EKS)** using **Helm charts**, enabling scalable, repeatable deployments and reducing release cycle times by 35%.

Cognizant Technology Solutions, Software Engineer

Jan 2021 – Jul 2021 | Hyderabad, India

- Developed and deployed key modules of an **inventory management platform** for Gap, using **Java, Spring Boot, Node.js, React, PostgreSQL, and Elasticsearch**, hosted on **Azure App Services** for scalability and resilience.
- Developed **React frontends** with TypeScript and Material UI, improving component performance and load time by 40%.
- Developed **Node.js backend services** with asynchronous event handling and **WebSocket communication**, enabling near real-time updates of inventory levels across warehouses and retail stores.
- Built and optimized **CI/CD pipelines** with **Jenkins**, automating build, test, and deployment of **RESTful** Microservices to cut release cycles from weeks to days and improve delivery reliability.

EDUCATION

University of Maryland Baltimore county,

Aug 2022 – May 2024 | Baltimore, USA

Master of Science in Computer Science (3.7/4.0)

Relevant coursework: Distributed Systems, Machine Learning, Cloud Computing, Data Structures & Algorithms.